

CSE 432 Robotics

Velocity Kinematics – The Manipulator Jacobian

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Introduction

- Forward kinematic equations define a function between the space of Cartesian positions and orientations and the space of joint positions.
- The velocity relationships are then determined by the Jacobian of this function.
- The Jacobian is a matrix that can be thought of as the vector version of the ordinary derivative of a scalar function.
- The Jacobian is one of the most important quantities in the analysis and control of robot motion.



ANGULAR VELOCITY: FIXED AXIS

- When a rigid body moves in a pure rotation about a fixed axis, every point of the body moves in a circle.
- The centers of these circles lie on the axis of rotation.
- As the body rotates, a perpendicular from any point of the body to the axis sweeps out an angle, and this angle is the same for every point of the body.
- If $\mathbf{k}$ is a unit vector in the direction of the axis of rotation, then the angular velocity is given by

$$\omega = \dot{\theta} \mathbf{k}$$

- in which $\dot{\theta}$ is the time derivative of θ .



ANGULAR VELOCITY: FIXED AXIS

- Given the angular velocity of the body, the linear velocity of any point on the body is given by the equation

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

- To specify the orientation of a rigid object, we attach a coordinate frame rigidly to the object, and then specify the orientation of the attached frame.
- Since every point on the object experiences the same angular velocity, and each point of the body is in a fixed geometric relationship to the body-attached frame, the angular velocity is a property of the attached coordinate frame itself.



SKEW SYMMETRIC MATRICES

- An $n \times n$ matrix S is said to be skew symmetric if and only if:

$$S^T + S = 0$$

- 3×3 skew symmetric matrices are denoted by $so(3)$. If $S \in so(3)$ has components $s_{ij}, i, j = 1, 2, 3$ then

$$s_{ij} + s_{ji} = 0, \quad i, j = 1, 2, 3$$
$$s_{ii} = 0$$

The diagonal terms of S are zero and the off-diagonal terms s_{ij} , $i \neq j$ satisfy $s_{ij} = -s_{ji}$.



Properties of Skew-Symmetric Matrices

- If $a = (a_x, a_y, a_z)$ is a 3-vector, the skew symmetric matrix $S(a)$ is defined as

$$S(a) = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$$

- For any vectors a and $p \in \mathbb{R}^3$

$$a \times p = S(a)p$$



Properties of Skew-Symmetric Matrices

- The operator S is linear, i.e.,

$$S(\alpha a + \beta b) = \alpha S(a) + \beta S(b)$$

- for any vectors $a, b \in \mathbb{R}^3$ and scalars α, β .
- For any rotation matrix R and any vector $a \in \mathbb{R}^3$:

$$R S(a) R^T = S(Ra)$$

- Equivalently:

$$RS(a) = S(Ra) R$$

- **This is because:**

- $S(x)$ represents the cross-product operation: $S(x)y = x \times y$
- Rotations preserve cross products: $R(x \times y) = (Rx) \times (Ry)$



The Derivative of a Rotation Matrix

- Since R is orthogonal for all θ it follows that

$$R(\theta)R(\theta)^T = I$$

- Differentiating both sides with respect to θ

$$\frac{dR}{d\theta}R(\theta)^T + R(\theta)\frac{dR^T}{d\theta} = 0$$

- Let us define the matrix S as

$$S := \frac{dR}{d\theta}R(\theta)^T$$



The Derivative of a Rotation Matrix

- Then

$$\frac{dR}{d\theta} = SR(\theta)$$

- Computing the derivative of the rotation matrix R is equivalent to a matrix multiplication by a skew symmetric matrix S .
- Similarly, Let $R_{k,\theta}$, be a rotation about the axis defined by k

$$\frac{dR_{k,\theta}}{d\theta} = S(k)R_{k,\theta}$$



Example

- For $R = R_{x,\theta}$ compute $\frac{dR}{d\theta}$
- Directly differentiate each element

$$\frac{dR_x(\theta)}{d\theta} = \frac{d}{d\theta} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\frac{dR_x(\theta)}{d\theta} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\sin \theta & -\cos \theta \\ 0 & \cos \theta & -\sin \theta \end{bmatrix}$$



Example

- For $R = R_{x,\theta}$ compute $\frac{dR}{d\theta}$: $\frac{dR_{x,\theta}}{d\theta} = S(i)R_{x,\theta}$

$$S(i) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\frac{dR}{d\theta} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$\frac{dR}{d\theta} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\sin \theta & -\cos \theta \\ 0 & \cos \theta & -\sin \theta \end{bmatrix}$$



ANGULAR VELOCITY: GENERAL Axis

- Suppose that a rotation matrix R is time varying, so that $R = R(t) \in SO(3)$ for every $t \in R$.
- If $R(t)$ is continuously differentiable as a function of t , then
$$\dot{R}(t) = S(\omega(t))R(t)$$
- Now, since $S(t)$ is skew symmetric, it can be represented as $S(\omega(t))$ for a unique vector $\omega(t)$.
- This vector $\omega(t)$ is the angular velocity of the rotating frame with respect to the fixed frame at time t .



Example

- Suppose that $R(t) = R_{x,\theta(t)}$. Compute $\dot{R}(t)$

- Use the chain rule:

$$\begin{aligned}\dot{R} &= \frac{dR}{dt} = \frac{dR}{d\theta} \frac{d\theta}{dt} = \dot{\theta} S(i) R(t) = S(\omega(t)) R(t) \\ \dot{R} &= \dot{\theta} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \\ \dot{R} &= \begin{bmatrix} 0 & -\sin \theta & -\cos \theta \\ 0 & \cos \theta & -\sin \theta \end{bmatrix} \dot{\theta}\end{aligned}$$

- Where

- $i = (1,0,0)$
- $\omega = i \dot{\theta}$ is the angular velocity



Addition of angular velocities

- Recalling that

$$R_2^0(t) = R_1^0(t) R_2^1(t)$$

- Differentiating:

$$\dot{R}_2^0 = \dot{R}_1^0 R_2^1 + R_1^0 \dot{R}_2^1$$

- Using

$$\dot{R}(t) = S(\omega(t))R(t),$$

- the left-hand side becomes:

$$\dot{R}_2^0 = S(\omega_{0,2}^0) R_2^0$$



Addition of angular velocities

$$\dot{R}_2^0 = \dot{R}_1^0 R_2^1 + R_1^0 \dot{R}_2^1$$

- **First Term**

$$\dot{R}_1^0 R_2^1 = S(\omega_{0,1}^0) R_1^0 R_2^1 = S(\omega_{0,1}^0) R_2^0$$

- **Second Term**

$$R_1^0 \dot{R}_2^1 = R_1^0 S(\omega_{1,2}^1) R_2^1 = S(R_1^0 \omega_{1,2}^1) R_2^0$$

- **Thus:**

$$S(\omega_{0,2}^0) R_2^0 = [S(\omega_{0,1}^0) + S(R_1^0 \omega_{1,2}^1)] R_2^0$$
$$\omega_{0,2}^0 = \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 = \omega_{0,1}^0 + \omega_{1,2}^0$$



Addition of angular velocities

- **Generalization:** For any number of frames:

$$R_n^0 = R_1^0 R_2^1 \cdots R_n^{n-1}$$

- Thus, the angular velocities can be added once they are expressed relative to the same coordinate frame, in this case $O_0x_0y_0z_0$.

$$\dot{R}_n^0 = S(\omega_{0,n}^0) R_n^0$$

- in which

$$\begin{aligned}\omega_{0,n}^0 &= \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 + R_2^0 \omega_{2,3}^2 + \cdots + R_{n-1}^0 \omega_{n-1,n}^{n-1} \\ \omega_{0,n}^0 &= \omega_{0,1}^0 + \omega_{1,2}^0 + \omega_{2,3}^0 + \cdots + \omega_{n-1,n}^0\end{aligned}$$



Linear velocity of a point attached to a rotating frame

- Suppose the point p is rigidly attached to frame $o_1x_1y_1z_1$, which is rotating relative to the frame $o_0x_0y_0z_0$.
- Then the coordinates of p with respect to the frame $o_0x_0y_0z_0$ are given by

$$p^0 = R_1^0 p^1$$

- The velocity $\dot{p}^0$ is then given by

$$\begin{aligned}\dot{p}^0 &= S(\omega^0) R_1^0 p^1 \\ \dot{p}^0 &= S(\omega^0) p^0 = \omega^0 \times p^0\end{aligned}$$



Linear velocity of a point attached to a moving frame

- Now suppose that the motion of the frame $o_1x_1y_1z_1$ relative to $o_0x_0y_0z_0$ is more general.
- Suppose that the homogeneous transformation relating the two frames is time-dependent, so that

$$H_1^0(t) = \begin{bmatrix} R_1^0(t) & o_1^0(t) \\ 0 & 1 \end{bmatrix}$$
$$p^0 = R_1^0 p^1 + o$$

- Differentiating the above expression

$$\dot{p}^0 = S(\omega^0)p^0 + \dot{o} = \omega^0 \times p^0 + v$$



DERIVATION OF THE JACOBIAN

- Consider an n-link manipulator with joint variables $q_1, q_2, \dots, q_n$.
- Let the transformation from the end-effector frame to the base frame

$$T_n^0(q) = \begin{bmatrix} R_n^0(t) & o_n^0(t) \\ 0 & 1 \end{bmatrix}$$

- Let ω_n^0 is the angular velocity vector of the end-effector, then

$$S(\omega_n^0) = \dot{R}_n^0 (R_n^0)^T$$

- Let the linear velocity of the end effector

$$v_n^0 = \dot{o}_n^0$$



DERIVATION OF THE JACOBIAN

- We seek expressions of the form

$$\begin{aligned}v_n^0 &= J_v \dot{q} \\ \omega_n^0 &= J_\omega \dot{q}\end{aligned}$$

where J_v and J_ω are $3 \times n$ matrices. We may write

$$\xi = J\dot{q},$$

$$\xi = \begin{bmatrix} v_n^0 \\ \omega_n^0 \end{bmatrix}, \quad J = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix}$$

- The vector ξ is sometimes called a body velocity.
- The matrix J is called the Manipulator Jacobian or Jacobian for short.
- Note that J is a $6 \times n$ matrix where n is the number of links.



Angular Velocity

- If the i^{th} joint is revolute, then the i^{th} joint variable $q_i = \theta_i$ and the axis of rotation is z_{i-1} .
- Let ω_i^{i-1} represent the angular velocity of link i that is imparted by the rotation of joint i , expressed relative to frame $O_{i-1}x_{i-1}y_{i-1}z_{i-1}$.
- This angular velocity is expressed in the frame $i - 1$ by
$$\omega_i^{i-1} = \dot{q}_i z_{i-1}^{i-1} = \dot{q}_i k, \quad k = [0 \quad 0 \quad 1]^T$$
- If the i^{th} joint is prismatic, then the motion of frame i relative to frame $i - 1$ is a translation, thus
$$\omega_i^{i-1} = 0$$



Angular Velocity

- The overall angular velocity of the end-effector, ω_n^0 :

$$\omega_n^0 = \rho_1 \dot{q}_1 k + \rho_2 \dot{q}_2 R_1^0 k + \cdots + \rho_n \dot{q}_n R_{n-1}^0 k = \sum_{i=1}^n \rho_i \dot{q}_i z_{i-1}^0$$

$$\rho_i = \begin{cases} 1 & \text{for revolute joint } i \\ 0 & \text{for prismatic joint } i \end{cases}$$

- Thus, the lower half of the Jacobian J_ω , is given as

$$J_\omega = [\rho_1 z_0 \cdots \rho_n z_{n-1}]$$

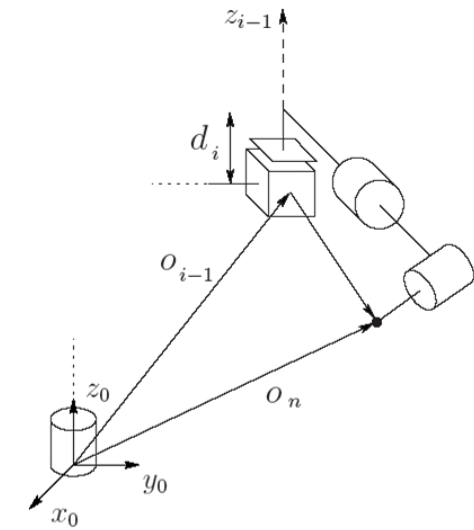
- Note that in this equation, we have omitted the superscripts for the unit vectors along the z -axes.



Linear Velocity

The linear velocity of the end-effector

$$\dot{o}_n^0 = \sum_{i=1}^n \frac{\partial o_n^0}{\partial q_i} \dot{q}_i$$



- Thus, we see that the i^{th} column of J_v , which we denote as J_{vi} is given by

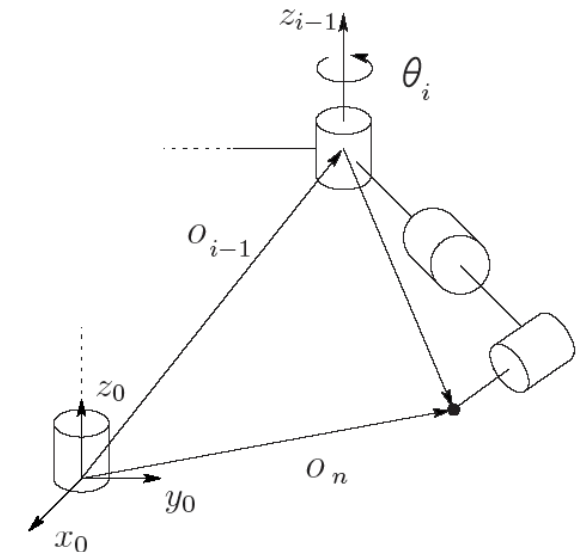
$$J_{vi} = \frac{\partial o_n^0}{\partial q_i}$$

(i) **Case 1: Prismatic Joints**

$$J_{vi} = z_{i-1}^0$$

(i) **Case 2: Revolute Joints**

$$J_{vi} = z_{i-1}^0 \times (o_n^0 - o_{i-1}^0)$$





Combining the Angular and Linear Jacobians

- The upper half of the Jacobian J_v is given as

$$J_v = [J_{v_1} \cdots J_{v_n}]$$

- where the i^{th} column J_{v_i} is

$$J_{v_i} = \begin{cases} z_{i-1}^0 \times (o_n^0 - o_{i-1}^0) & \text{for revolute joint } i \\ z_{i-1}^0 & \text{for prismatic joint } i \end{cases}$$

- The lower half of the Jacobian is given as

$$J_\omega = [J_{\omega_1} \cdots J_{\omega_n}]$$

- where the i^{th} column J_{ω_i} is

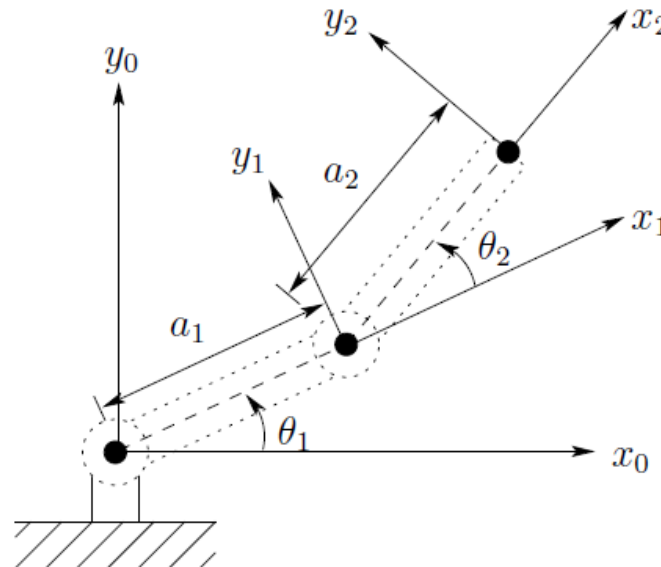
$$J_{\omega_i} = \begin{cases} z_{i-1}^0 & \text{for revolute joint } i \\ 0 & \text{for prismatic joint } i \end{cases}$$



Example Two-Link Planar Manipulator

- Consider the two-link planar manipulator. Since both joints are revolute the Jacobian matrix, which in this case is 6×2 , is of the form

$$J(q) = \begin{bmatrix} z_0 \times (o_2 - o_0) & z_1 \times (o_2 - o_1) \\ z_0 & z_1 \end{bmatrix}$$





Example Two-Link Planar Manipulator

- The various quantities above are easily seen to be

$$o_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, o_1 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \end{bmatrix}, o_2 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \\ 0 \end{bmatrix}, \quad z_0 = z_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

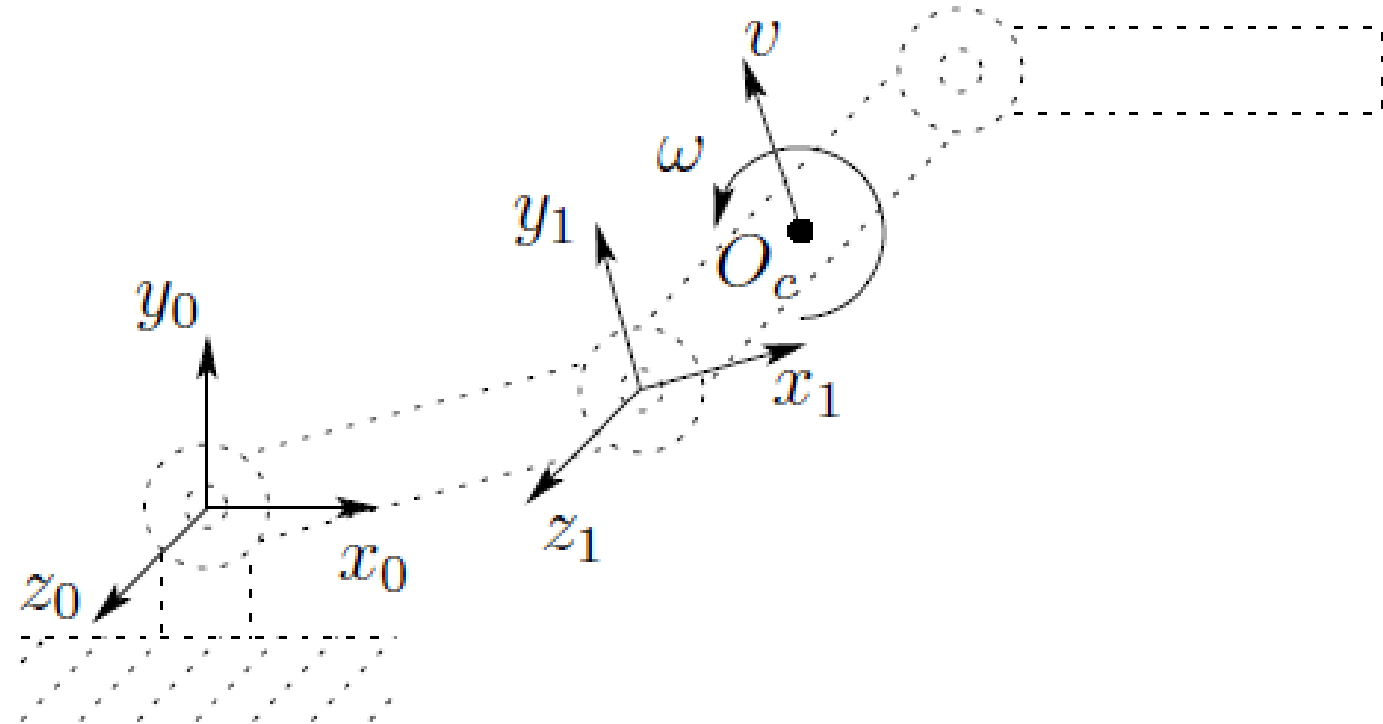
- Performing the required calculations then yields

$$J = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$



Example Three-Link Planar Manipulator

- Consider the three-link planar manipulator. Suppose we wish to compute the linear velocity v and the angular velocity ω of the center of link 2 as shown.





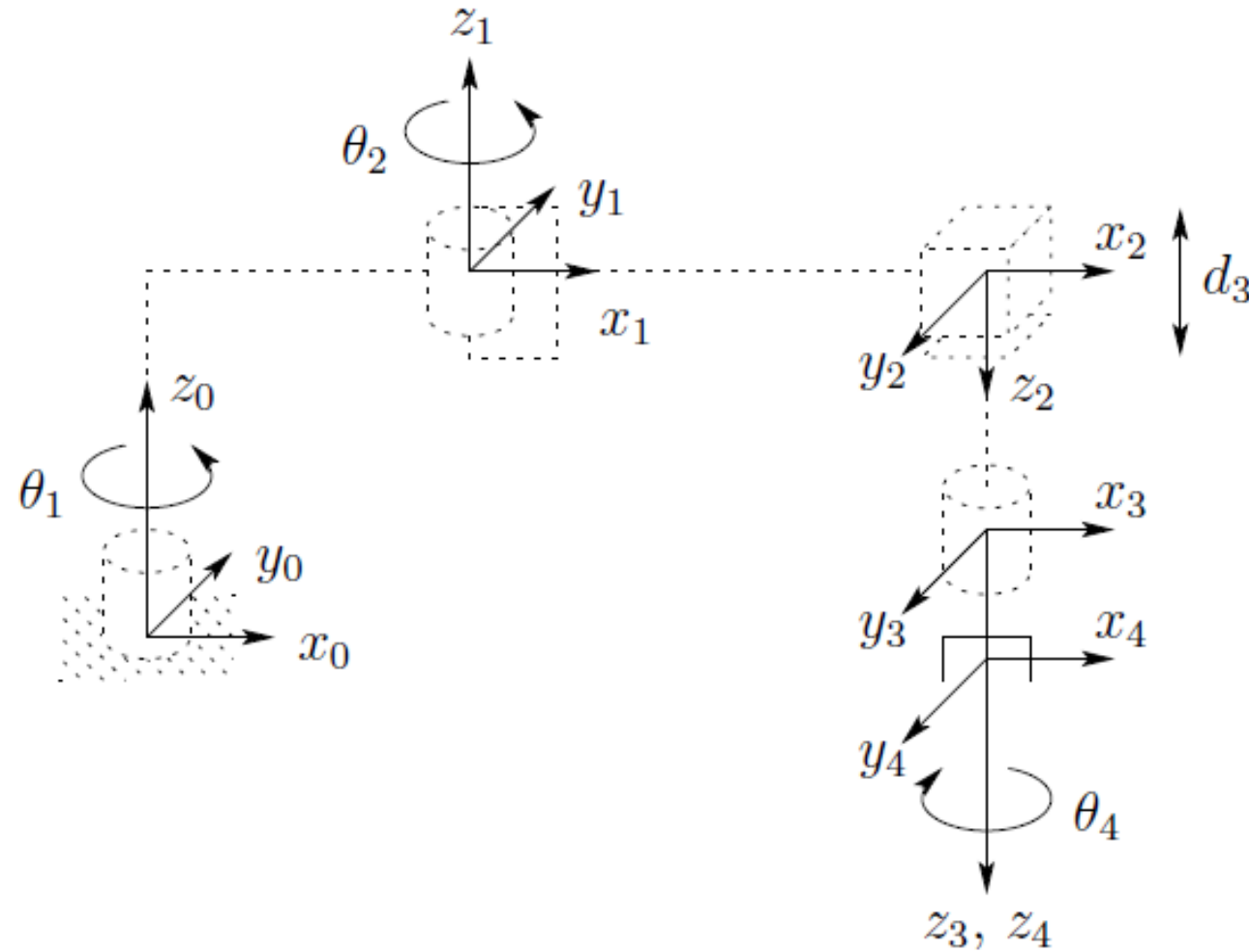
Example Three-Link Planar Manipulator

- In this case we have that

$$J(q) = \begin{bmatrix} z_0 \times (o_c - o_0) & z_1 \times (o_c - o_1) & 0 \\ z_0 & z_1 & 0 \end{bmatrix}$$



Example SCARA Manipulator





Example SCARA Manipulator

$$o_1 = \begin{bmatrix} a_1 c_1 \\ a_1 s_1 \\ 0 \end{bmatrix}, o_2 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \\ 0 \end{bmatrix}, o_4 = \begin{bmatrix} a_1 c_1 + a_2 c_{12} \\ a_1 s_1 + a_2 s_{12} \\ d_3 - d_4 \end{bmatrix}$$

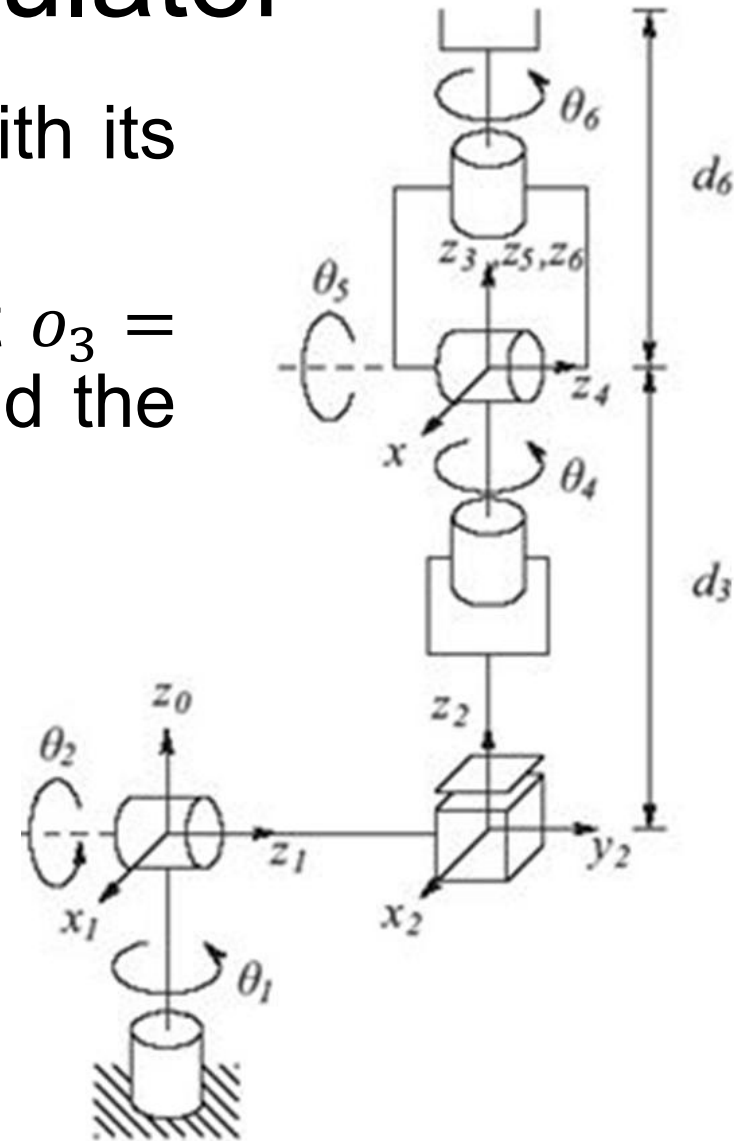
- Similarly, $z_0 = z_1 = [0 \ 0 \ 1]^T$, and $z_2 = z_3 = [0 \ 0 \ -1]^T$.
- The Jacobian of the SCARA Manipulator is

$$J = \begin{bmatrix} -a_1 s_1 - a_2 s_{12} & -a_2 s_{12} & 0 & 0 \\ a_1 c_1 + a_2 c_{12} & a_2 c_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix}$$



Example Stanford Manipulator

- Consider the Stanford manipulator with its associated DH-coordinate frames.
- Note that joint 3 is prismatic and that $o_3 = o_4 = o_5$ own of the spherical wrist and the frame assignment.
- Denoting this common origin by o .





Example Stanford Manipulator

- we see that the columns of the Jacobian have the form

$$J_i = \begin{bmatrix} z_{i-1} \times (o_6 - o_{i-1}) \\ z_{i-1} \end{bmatrix} \quad i = 1, 2$$

$$J_3 = \begin{bmatrix} z_2 \\ 0 \end{bmatrix}$$

$$J_i = \begin{bmatrix} z_{i-1} \times (o_6 - o) \\ z_{i-1} \end{bmatrix} \quad i = 4, 5, 6$$

$$o_6 = \begin{bmatrix} c_1 s_2 d_3 - s_1 d_2 + d_6 (c_1 c_2 c_4 s_5 + c_1 c_5 s_2 - s_1 s_4 s_5) \\ s_1 s_2 d_3 - c_1 d_2 + d_6 (c_1 s_4 s_5 + c_2 c_4 s_1 s_5 + c_5 s_1 s_2) \\ c_2 d_3 + d_6 (c_2 c_5 - c_4 s_2 s_5) \end{bmatrix}$$

$$o_3 = \begin{bmatrix} c_1 s_2 d_3 - s_1 d_2 \\ s_1 s_2 d_3 + c_1 d_2 \\ c_2 d_3 \end{bmatrix}$$



Example Stanford Manipulator

$$\begin{aligned} z_0 &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, & z_1 &= \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \\ z_2 &= \begin{bmatrix} c_1 s_2 \\ s_1 s_2 \\ c_2 \end{bmatrix}, & z_3 &= \begin{bmatrix} -c_1 s_2 \\ s_1 s_2 \\ c_2 \end{bmatrix}, \\ z_4 &= \begin{bmatrix} -c_1 c_2 s_4 - s_1 c_4 \\ -s_1 c_2 s_4 + c_1 c_4 \\ s_2 s_4 \end{bmatrix} \\ z_5 &= \begin{bmatrix} c_1 c_2 c_4 s_5 - s_1 s_4 s_5 + c_1 s_2 c_5 \\ s_1 c_2 c_4 s_5 + c_1 s_4 s_5 + s_1 s_2 c_5 \\ -s_2 c_4 s_5 + c_2 c_5 \end{bmatrix} \end{aligned}$$



Any Questions